

1. What is the Difference Between Live Connection and Extract Connection?

Ans:

In Tableau, data can be connected using either a **Live Connection** or an **Extract Connection**.

Live Connection

Connects directly to the original data source.
Always displays the latest data from the source.
Requires a connection to the source database.
Performance depends on database speed and network.
Suitable for real-time reporting and monitoring.

Extract Connection

Stores a snapshot (copy) of the data in Tableau.
Displays data as of the last extract refresh.
Can be used offline after extraction.
Generally faster because data is stored locally.
Suitable for historical analysis and improved performance.

Example:

A hospital wants to monitor emergency room admissions in real time.

- **Live Connection:** Shows the latest admissions as soon as they are entered into the database.
- **Extract Connection:** Shows admissions data only up to the last refresh time.

2. What is the Difference Between Data Blending and Joining in Tableau?

Ans:

Both Data Blending and Joining are used to combine data from multiple sources, but they work differently.

Data Blending

Combines data from multiple data sources.
Uses a Primary and Secondary data source.
Data is combined after aggregation.
Relationships are based on linking fields.
Useful when data sources cannot be directly joined.
Generally slower for large datasets.

Joining

Combines tables before analysis.
Uses tables from the same or compatible data source.
Data is combined at the row level.
Relationships are based on join keys and join types.
Useful when tables share a common key.
Usually faster and more efficient.

Example of Joining:

Table A: Patients

Patient ID Name

101	John
102	Mary

Table B: Admissions

Patient ID Admission Date

101	01-Jan-2025
102	05-Jan-2025

Joining on **Patient ID** combines the tables into a single dataset.

Example of Data Blending:

- Data Source 1: Hospital Admissions (Excel)
- Data Source 2: Patient Satisfaction Scores (SQL Server)

If these sources cannot be directly joined, Tableau can blend them using a common field such as **Department**.